

PES University — PES1UG24AM205

SE Lab 1: Requirements Engineering & UML Use-Case Modelling

Scenario #40 | Retail, E-Commerce & Finance
Student Startup Crowdfunding Platform

Deliverable 1: Functional Requirements (FR-001 to FR-005)

ID	Description	Priority	Acceptance Criteria
FR-001	The system shall allow a Student Founder to create a crowdfunding campaign by defining a project title, description, funding goal, milestone-based tranches, and reward tiers.	High	Pass: Campaign is created with all mandatory fields, unique ID generated, and status set to "Pending Approval". Fail: Campaign created without funding goal or milestone details.
FR-002	The system shall collect backer pledges and disburse funds to the student venture in tranches only upon verification of predetermined project milestones by the Faculty Committee.	High	Pass: Milestone evidence is reviewed, approved by Faculty, and the corresponding tranche is released to the founder's verified account. Fail: Campaign funds are disbursed without milestone verification or faculty approval.
FR-003	The system shall enable a Campaign Backer to browse active campaigns, view milestone progress, and select a reward tier before pledging a monetary amount.	Medium	Pass: Backer successfully selects a tier, pledge amount is recorded, and a confirmation receipt is generated. Fail: Pledge is recorded without tier selection or campaign status is not "Active".
FR-004	The system shall allow a Student Founder to submit milestone evidence (documents, images, demo links) and track the review status in real time.	Medium	Pass: Evidence is uploaded, timestamped, notification sent to Faculty Committee, and status updates to "Under Review". Fail: Evidence submission accepted without file validation or no notification triggered.
FR-005	The system shall automatically notify all stakeholders (Founder, Backers, Faculty) via email/push when a milestone is verified, a tranche is released, or a campaign reaches its funding goal.	Low	Pass: Relevant stakeholders receive timely notifications with accurate campaign/tranche details. Fail: Notifications are missed, delayed beyond 5 minutes, or sent to wrong recipients.

Deliverable 1: Non-Functional Requirements (NFR-001 & NFR-002)

ID	Type	Description	Priority	Acceptance Criteria
NFR-001	Performance & Security	The payment processing gateway must comply with PCI-DSS guidelines with tokenized payment authorizations. All financial transactions must complete within ≤ 3 seconds under normal load and ≤ 5 seconds under simulated peak load (500 concurrent transactions).	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load; penetration test reveals zero critical vulnerabilities; tokenization verified in database logs. Fail: Transaction latency exceeds thresholds, plain-text card data is stored, or PCI-DSS audit fails.
NFR-002	Availability & Reliability	The platform shall maintain 99.5% uptime during academic term periods, with automated daily backups of all campaign data, pledge records, and milestone evidence. Recovery Time Objective (RTO) must not exceed 4 hours.	High	Pass: Uptime monitoring logs show $\geq 99.5\%$ availability over a 30-day period; backup restoration test completes within 4 hours with zero data loss. Fail: Unplanned downtime exceeds 0.5% in a month, backup restoration fails, or data loss is detected.

Deliverable 2: UML Use-Case Diagram

The following diagram models all actors, primary use cases, and includes at least one «include» and one «extend» relationship.



Diagram Elements Summary

Element	Details
Actors	Student Founder, Campaign Backer, Faculty Committee
System Boundary	Crowdfunding Platform
Primary Use Cases	Create Campaign, Submit Milestone Evidence, Manage Campaign, Browse Campaigns, Pledge & Select Reward Tier, Track Campaign Progress, Verify Milestone & Release Tranche, Process Secure Payment
«include»	Create Campaign → Process Secure Payment (platform fee/verification); Pledge & Select Reward Tier → Process Secure Payment (mandatory payment flow)
«extend»	Submit Milestone Evidence → Verify Milestone & Release Tranche (evidence submission triggers the verification extension)

Deliverable 3: Use-Case Flow Specification

UC-002: Pledge & Select Reward Tier

Field	Specification
Use-Case ID	UC-002
Use-Case Name	Pledge & Select Reward Tier
Actor(s)	Primary: Campaign Backer; Supporting: Process Secure Payment (system)
Description	Allows a backer to browse an active campaign, select a reward tier, enter a pledge amount, and complete a secure payment. The system records the pledge and issues a digital receipt.

Preconditions:

1. Backer is authenticated and logged into the platform.
2. Target campaign status is "Active" and has not reached its funding deadline.
3. Backer has a valid payment method linked to their profile.

Postconditions:

Success: Pledge is recorded, payment is authorized, backer receives confirmation, and campaign's pledged total is updated.

Failure: Payment is declined, pledge is not recorded, and backer is notified with error details.

Main Success Scenario

Step	Actor Action	System Response
1	Backer navigates to an active campaign page and clicks "Back This Project".	System displays the campaign's reward tiers with descriptions, minimum pledge amounts, and estimated delivery dates.
2	Backer selects a reward tier (or chooses "No Reward") and enters a custom pledge amount $\geq$ the tier minimum.	System validates the amount and enables the "Proceed to Payment" button.
3	Backer reviews the pledge summary and clicks "Proceed to Payment".	System redirects to the secure PCI-DSS compliant checkout page with tokenized payment fields.
4	Backer enters/confirm payment details and clicks "Confirm Pledge".	System invokes Process Secure Payment («include»). Payment gateway tokenizes card data, authorizes the transaction, and returns a success token.
5	—	System records the pledge, updates the campaign's total pledged amount, locks the reward tier inventory (if limited), generates a unique pledge ID, and sends a confirmation email + in-app receipt to the backer.
6	—	System displays the "Pledge Successful" screen with pledge ID, reward tier details, and a link to track campaign progress.

Alternate Flow: A1 — Payment Authorization Failure (at Step 4)

Step	Actor Action	System Response
A1.1	Backer submits payment details.	Payment gateway returns a decline code (e.g.,

		insufficient funds, expired card, fraud flag).
A1.2	—	System does NOT record the pledge. It logs the failed attempt for audit purposes, displays a user-friendly error message ("Payment could not be processed. Please check your details or try a different payment method."), and prompts the backer to retry or cancel.
A1.3	Backer may retry with corrected details or click "Cancel".	If retry: return to Step 4. If cancel: system aborts the flow, releases any temporary holds, and returns the backer to the campaign page with no pledge recorded.

